ENEE12015 - Electrical Power Engineering

Lecture Notes

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1 Introduction to Electrical Power Systems

1.1 Early Development:

1800's - Electricity and Magnetism well understood intellectually
1831 - Magnetica field was discovered to generate electricity
1845 - First commerical operations of telegraph
1860 - First electrical generator
1870 - First incandesent light bulb
1882 - First commericial power system in NY City
1883 - Polyphase induction motor
1885 - Commericial availability of AC power

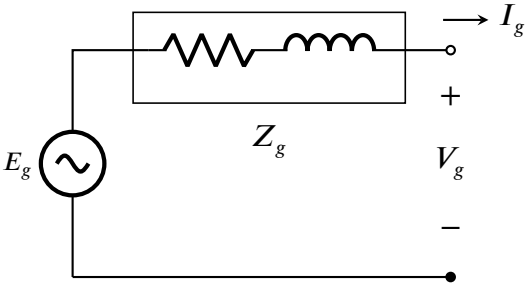
1.2 Power System Compenets:

- Generators
- Transformers
- Transmission lines
- Substations/Terminal Stations
- Distribution lines
- Electrical loads

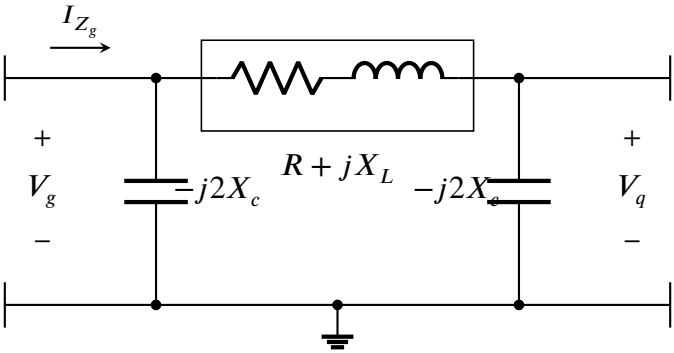
When transmitting power, the best senario is high voltage, low current transmission. Ideally below the resistivity of the power lines.

1.3 Power System Models:

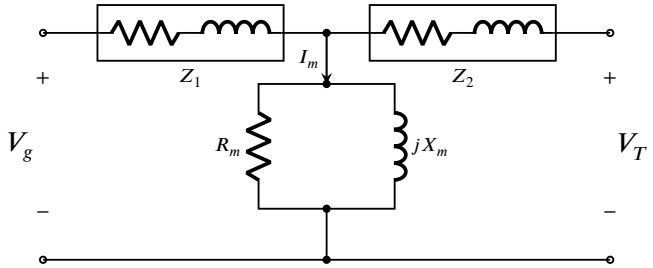
1.3.1 Generators



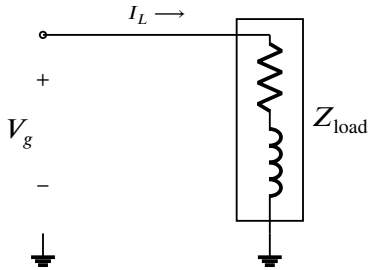
1.3.2 Transmission/Distribution Lines



1.3.3 Transformers



1.3.4 Loads



1.4 Basic Theory:

Ohm's Law - $V = IZ$
KVL - $\sum V = 0$

KCL - $\sum I_{in} = \sum I_{out}$
Power Flow - $S = P + jQ = VI^*$

1.5 Current and Power Flow:

- Technically current flow within an AC system is arbitrary
- Power is clear from source to load
- Steady state is only assumed

1.6 Power in AC Circuits:

Average Power over time period T

$$P = \frac{1}{T}\int_{t_0}^{t_0+T} v(t)i(t)dt$$

Instantaneous Power

$$p(t) = v(t)i(t)$$

When a circuit is in steady state, all voltages and currents become sinusodial. The periodic nature of the waveforms enables more specific derivation of power relationships.

$$P_{av} = \frac{E_m I_m}{2} = \frac{E_m}{\sqrt{2}} \times \frac{I_m}{\sqrt{2}} = E_{RMS} I_{RMS}$$

This only works when $v(t)$ and $i(t)$ are in phase. When a small phase shift occurs:

$$P_{av} = \frac{E_m I_m}{2}\cos(\theta) = E_{RMS} I_{RMS}\cos(\theta)$$

When $i(t)$ lags $v(t)$ by 90°

$$P_{av} = E_{RMS} I_{RMS}\cos(90°) = 0$$

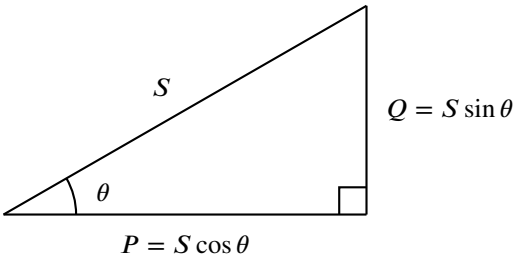
1.7 Reactive Power:

Power that dose no useful work within a system.

$$Q = E_{RMS} I_{RMS}\sin(\phi))$$

VAR is a defined value and does not relate to the area under P. When $v(t0$ and $i(t)$ are in phase ($\phi = 0$), $Q = 0$. When reactive power is maximum ($\phi = 90°$), $Q = E_{RMS} I_{RMS}$ and $P_{av} = 0$.

1.8 Power Triangles:



Q is wished to be minised, allowing the maximum S to used within the load (active power) usefully.

1.9 Power Flow Direction:

A positive P present in a power triangle indicates that ϕ is between $-90°$ and $90°$. With a negative P show ϕ to be $> 90°$. It is necessary to assign a current direction in AC circuits, for the purpose of writing equations in a consistent manner. $Q > 0$, flowing in the same direction as current is assumed to go. $Q < 0$, flowing apposed to the assumed current direction.
Power Factor:

$$pf = \frac{P}{S} = \cos(\phi)$$

2 Three Phase Power

2.1 What is it?:

A 3-phase power system is just a connecting of 3 singlephase systems together for set operation. Each phase must operate at the same voltage and current magnitudes seperated by $120°$ phase angle.

2.2 Balanced Operating Conditions:

Under normal operating conditions all voltages, current, $120°$ angle shift and loads contain same characteristics.

2.3 Analysis of 3-Phase:

Assuming a system is under balanced conditions, means that only a single phase system needs to be analysed. Faults, abnormal load conditions, cause a system

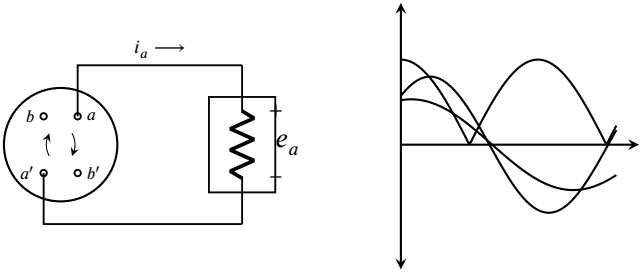
to become unbalanced.

2.4 Popularity of Polyphase Systems:

Polyphase machines can be designed to operate under constant total power conditions. Hence, these machines have smoother operations. These machines are then simpler to design, how all as tansmission lines can delivers greater power for less cost.

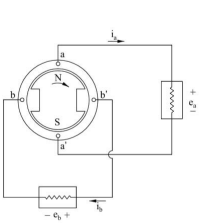
2.5 Power in Polyphase Systems:

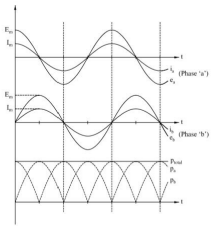
Single Phase: Resistive



$$
\begin{aligned}
p_a &= e_a i_a \\
&= E_m I_m \cos^2(\omega t) \\
P_{av} &= E_{RMS} I_{RMS}
\end{aligned}
$$

2-Phase: Resistive





Voltage induced in b-b' lags voltage in a-a' by 90 electrical degrees. In 2-pole machines, 1 electrical degree is equivalent to 1 mechanical degree.

$$\begin{aligned} e_a &= E_m \cos(\omega t) \\ i_a &= I_m \cos(\omega t) \end{aligned}$$

$$\begin{aligned} e_b &= E_m \cos(\omega t - 90°) \\ &= E_m \sin(\omega t) \\ i_b &= I_m \cos(\omega t - 90°) \\ &= I_m \sin(\omega t) \end{aligned}$$

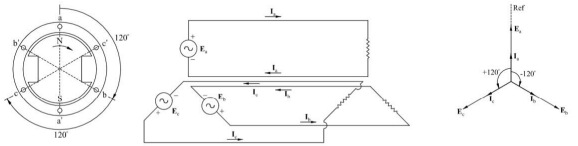
$$\begin{aligned} P_{Tot} &= P_a + P_b \\ &= E_m I_m \cos^2(\omega t) + E_m I_m \sin^2(\omega t) \\ &= E_m I_m \end{aligned}$$

There are two seperate loads. This P_{Tot} is true as long as the load is the same.

$$\begin{aligned} P_{av} &= P_{Tot} \\ &= 2E_{RMS} I_{RMS} \end{aligned}$$

$$\begin{aligned} S_{Tot} &= S_a + S_b \\ &= E_a I_a^* + E_b I_b^* \\ &= 2EI\angle 0° \end{aligned}$$

3-Phase: Resistive



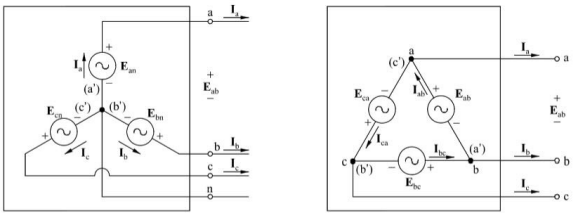
$$
\begin{aligned}
e_a &= E_m \cos(\omega t) \\
i_a &= I_m \cos(\omega t)
\end{aligned}
$$

$$
\begin{aligned}
e_b &= E_m \cos(\omega t - 120°) \\
i_b &= I_m \cos(\omega t - 120°)
\end{aligned}
$$

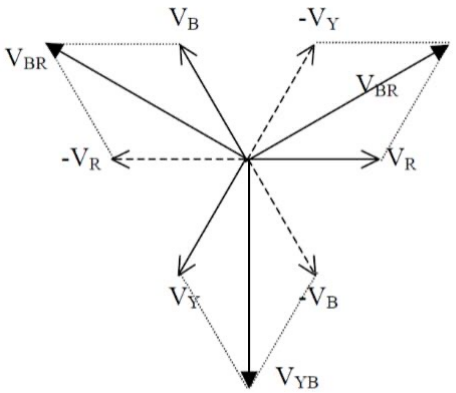
$$
\begin{aligned}
e_c &= E_m \cos(\omega t - 240°) \\
&= E_m \cos(\omega t + 120°) \\
i_c &= I_m \cos(\omega t - 240°) \\
&= I_m \cos(\omega t + 120°)
\end{aligned}
$$

$$
\begin{aligned}
P_{Tot} &= P_a + P_b + P_c \\
&= 1.5 E_m I_m \\
&= 3 E_{RMS} I_{RMS}
\end{aligned}
$$

2.6 3-Phase Connections:



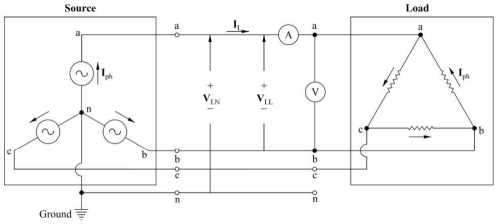
2.7 Balanced Systems:



- Angles between phases is 120°
- Magnitudes are all equal
- V_L would be 30° leading nearest phase voltage
- $|V_L|$ is $\sqrt{3}|V_P|$

$$I_L = I_P$$
$$|V_L| = \sqrt{3}|V_P|$$
$$|I_L| = \sqrt{3}|I_P|$$

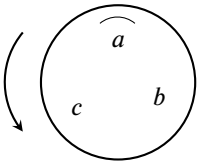
2.8 3-Phase Device Connection:



The delta-star and star-delta conversions can be done. The only requirement is that the device voltage ratings must be compatitable.

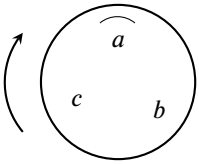
2.9 Phase Sequence:

The phase sequence denotes which phase becomes succesively positive.



+ve seq

$-a-b-c-a-b-c-$



–ve seq

$-a-c-b-a-c-b-$

2.10 3-Phase Solutions:

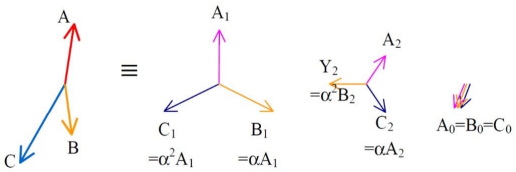
- Most 3-Phase system can be though of as balanced
- To solve an unbalanced 3-phase system is quite complex

2.11 Symmetrical Component Analysis:

Any unbalanced system can be split up into a series of balanced systems.

- Same phase sequence (Positive)
- Opposite phase sequence (Negative)
- In phase quantities (Zero)

2.12 Unbalanced Decomposition:

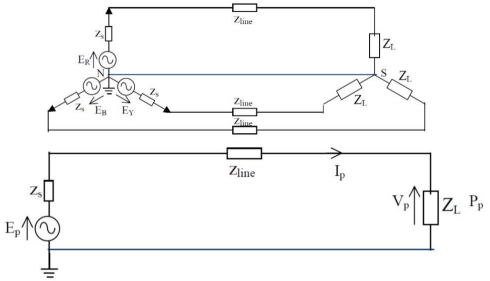


$$\alpha = 1\angle 120^\circ$$

$$A = A_0 + A_1 + A_2 = A_0 + A_1 + A_2$$
$$B = B_0 + B_1 + B_2 = A_0 + \alpha^2 A_1 + \alpha A_2$$
$$C = C_0 + C_1 + C_2 = A_0 + \alpha A_1 + \alpha^2 A_2$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha^2 & \alpha \\ 1 & \alpha & \alpha^2 \end{bmatrix} \begin{bmatrix} A_0 \\ A_1 \\ A_2 \end{bmatrix}$$

2.13 Balanced Analysis:



2.13.1 Wye Connection

$$I_P = I_L$$
$$V_P = \frac{V_L}{\sqrt{3}}$$

2.13.2 Delta Connection

$$V_P = V_L$$
$$I_P = \frac{I_L}{\sqrt{3}}$$

2.13.3 All

$$S_{3P} = 3V_P I_P = \sqrt{3} V_L I_L$$
$$P_{1P} = \frac{P_{3P}}{3}$$

2.14 Selecting Number of Phases:

The number of phases for a power system is determined by transmission system economies. That is done by analyzing the amount of power transferred per conductor. With the assumption that the size of the return conductor is equal to that phase conductor.

System	Power Rating	No. Conductors	Power per Conductor
1-Phase	P	2	0.5P
2-Phase	2P	3	0.667P
3-Phase	3P	4	0.75P
4-Phase	4P	5	0.8P